YOLOv8n Fine-Tuning on LISA Traffic Signs

This notebook walks through every step of fine-tuning a YOLOv8n model on the LISA traffic-sign dataset.

Steps

1. Install dependencies
2. Set configuration
3. Prepare the LISA dataset (CSV → YOLO format)
4. Inspect the dataset
5. Load the base model
6. Train
7. Review metrics
8. Save the best model to `trained_models/`
9. Run a quick validation inference

```
from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

```
import zipfile
from pathlib import Path

LISA_ZIP  = Path('/content/drive/MyDrive/OmniAI/Lisa.zip')
LISA_ROOT = Path('/content/lisa_data')

if LISA_ROOT.exists() and any(LISA_ROOT.iterdir()):
    print('Already extracted  — skipping')
else:
    print('Extracting Lisa.zip to local VM...')
    LISA_ROOT.mkdir(parents=True, exist_ok=True)
    with zipfile.ZipFile(LISA_ZIP, 'r') as z:
        z.extractall(LISA_ROOT)
```

```
    print('Extracted ')

# Verify structure
csv_files = list(LISA_ROOT.rglob('frameAnnotationsBOX.csv'))
print(f'frameAnnotationsBOX.csv files: {len(csv_files)}')
print('\nFirst 3:')
for f in csv_files[:3]:
    print(f'  {f}')
```

```
Extracting Lisa.zip to local VM...
Extracted
frameAnnotationsBOX.csv files: 24

First 3:
  /content/lisa_data/sample-nightClip1/sample-nightClip1/frameAnnotationsBOX.csv
  /content/lisa_data/sample-dayClip6/sample-dayClip6/frameAnnotationsBOX.csv
  /content/lisa_data/Annotations/Annotations/daySequence1/frameAnnotationsBOX.csv
```

Step 0 — Google Colab Setup

Run this cell **only on Colab**. Skip it if running locally.

```
import sys, os

IN_COLAB = 'google.colab' in sys.modules or 'COLAB_GPU' in os.environ

if IN_COLAB:
    # ── Mount Google Drive ──────────────────────────────────────────
    from google.colab import drive
    drive.mount('/content/drive')

    # ── Install deps ────────────────────────────────────────────────
    os.system('pip install ultralytics opencv-python-headless -q')

    # ── GPU check ───────────────────────────────────────────────────
    import torch
    gpu_name = torch.cuda.get_device_name(0) if torch.cuda.is_available() else 'None'
    vram_gb  = torch.cuda.get_device_properties(0).total_memory / 1e9 if torch.cuda.is_available() else 0
```

```
print(f'GPU : {gpu_name}  |  VRAM: {vram_gb:.1f} GB')

if not torch.cuda.is_available():
    raise RuntimeError('No GPU detected. Runtime → Change runtime type → A100 or V100 → Save & Reconnect'

# ── Batch size based on VRAM ──────────────────────────────────────────
# A100 = ~40GB, V100 = ~16GB, T4 = ~15GB
if vram_gb >= 35:       # A100
    _batch, _workers = 64, 8
elif vram_gb >= 14:     # V100 / T4
    _batch, _workers = 32, 4
else:
    _batch, _workers = 16, 2



# ── Path overrides ────────────────────────────────────────────────────
DRIVE_PROJECT = '/content/drive/MyDrive/OmniAI/training'

import pathlib
REPO_ROOT  = pathlib.Path(DRIVE_PROJECT)
LISA_ROOT  = pathlib.Path('/content/lisa_data')
YOLO_DATA  = pathlib.Path('/content/lisa_yolo')
BASE_MODEL = pathlib.Path('/content/drive/MyDrive/OmniAI/training_models/yolov8n_lisa_best.pt')
OUT_DIR    = pathlib.Path('/content/drive/MyDrive/OmniAI/training_models')

# ── Training params ───────────────────────────────────────────────────
EPOCHS   = 50
IMG_SIZE = 640
BATCH    = _batch
WORKERS  = _workers
LR0      = 0.001
PATIENCE = 20
AUGMENT  = True
FREEZE   = 0
DEVICE   = '0'

RUN_PROJECT = '/content/runs/train'
RUN_NAME    = 'lisa_yolov8n_v1.1'
VAL_SPLIT   = 0.15
SEED        = 42
```

```
    sys.path.insert(0, str(REPO_ROOT / 'training'))

    print(f'REPO_ROOT  : {REPO_ROOT}  exists={REPO_ROOT.exists()}')
    print(f'LISA_ROOT  : {LISA_ROOT}  exists={LISA_ROOT.exists()}')
    print(f'BASE_MODEL : {BASE_MODEL}  exists={BASE_MODEL.exists()}')
    print(f'DEVICE     : {DEVICE}  |  BATCH={BATCH}  WORKERS={WORKERS}  EPOCHS={EPOCHS}')
else:
    print('Not on Colab — skipping. Proceed to Step 2.')
```

```
Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", fo
GPU : Tesla T4  |  VRAM: 15.6 GB
REPO_ROOT  : /content/drive/MyDrive/OmniAI/training  exists=True
LISA_ROOT  : /content/lisa_data  exists=True
BASE_MODEL : /content/drive/MyDrive/OmniAI/training_models/yolov8n_lisa_best.pt  exists=True
DEVICE     : 0  |  BATCH=32  WORKERS=4  EPOCHS=50
```

```
# ── GPU Verification (Colab Pro) ──────────────────────────────────────────
import torch

print('torch version  :', torch.__version__)
print('CUDA available :', torch.cuda.is_available())

if torch.cuda.is_available():
    props = torch.cuda.get_device_properties(0)
    print(f'GPU name       : {props.name}')
    print(f'VRAM           : {props.total_memory / 1e9:.1f} GB')
    print(f'CUDA version   : {torch.version.cuda}')
    # Quick tensor test
    x = torch.zeros(1).cuda()
    print(f'Tensor device  : {x.device}')
    print('\nGPU is ready.')
else:
    print('\nNo GPU — go to: Runtime → Change runtime type → A100 → Save')
```

```
torch version  : 2.10.0+cu128
CUDA available : True
GPU name       : Tesla T4
```

```
VRAM           : 15.6 GB
CUDA version   : 12.8
Tensor device  : cuda:0

GPU is ready.
```

Step 1 — Install Dependencies

```
# Run once; restart kernel if packages were freshly installed
%pip install ultralytics opencv-python-headless --quiet
```

Step 2 — Configuration

Edit the variables below before running the rest of the notebook.

```
import sys, os
import torch
from pathlib import Path

# On Colab, Step 0 already set all variables — skip this cell
IN_COLAB = 'google.colab' in sys.modules or 'COLAB_GPU' in os.environ
if IN_COLAB:
    print('Colab detected — Step 0 config is active. Skipping Step 2.')
else:
    # ── Paths ──────────────────────────────────────────────────────

    REPO_ROOT  = pathlib.Path(DRIVE_PROJECT)
    LISA_ROOT  = pathlib.Path('/content/lisa_data')
    YOLO_DATA  = pathlib.Path('/content/lisa_yolo')
    BASE_MODEL = pathlib.Path('/content/drive/MyDrive/OmniAI/training_models/yolov8n_lisa_best.pt')
    OUT_DIR    = pathlib.Path('/content/drive/MyDrive/OmniAI/training_models')

    # ── Device — validate before trusting ──────────────────────────
    def _detect_device() -> str:
        if torch.cuda.is_available():
            try:
```

```
            torch.zeros(1).cuda()
            return '0'
        except Exception:
            pass
    if hasattr(torch.backends, 'mps') and torch.backends.mps.is_available():
        try:
            torch.zeros(1).to('mps')
            return 'mps'
        except Exception:
            pass
    return 'cpu'

DEVICE = _detect_device()

# ── Training hyper-parameters ──────────────────────────────────────────
EPOCHS   = 50
IMG_SIZE = 640
BATCH    = 8  if DEVICE == 'cpu' else 16
WORKERS  = 0  if DEVICE == 'cpu' else 4
LR0      = 0.001
PATIENCE = 15
AUGMENT  = True
FREEZE   = 0

RUN_PROJECT = str(REPO_ROOT / 'runs' / 'train')
RUN_NAME    = 'lisa_yolov8n'
VAL_SPLIT   = 0.15
SEED        = 42

sys.path.insert(0, str(REPO_ROOT / 'training'))

print('REPO_ROOT :', REPO_ROOT)
print('LISA_ROOT :', LISA_ROOT, '→ exists:', LISA_ROOT.exists())
print('BASE_MODEL:', BASE_MODEL, '→ exists:', BASE_MODEL.exists())
print(f'Device    : {DEVICE}')
```

```
Colab detected — Step 0 config is active. Skipping Step 2.
```

Step 3 — Prepare the LISA Dataset

Reads all `frameAnnotationsBOX.csv` files, converts bounding boxes to normalised YOLO format, splits into train / val, and writes `dataset.yaml`.

> **Skip this cell** if you have already run it once and `lisa_yolo/` exists.

```
import sys
from pathlib import Path

# LISA already extracted to /content/lisa_data
LISA_ROOT = Path('/content/lisa_data')

# Add training dir to path for imports
sys.path.insert(0, str(REPO_ROOT))

from prepare_lisa import prepare

if YOLO_DATA.exists() and any(YOLO_DATA.iterdir()):
    print('LISA already prepared  — skipping')
    yaml_path = YOLO_DATA / 'dataset.yaml'
else:
    print('Preparing LISA dataset...')
    yaml_path = prepare(
        lisa_root  = str(LISA_ROOT),
        out_root   = str(YOLO_DATA),
        val_split  = VAL_SPLIT,
        seed       = SEED,
        max_images = None,    # ← use ALL images
    )

print(f'\ndataset.yaml: {yaml_path}')
```

```
Preparing LISA dataset...

dataset.yaml: /content/lisa_yolo/dataset.yaml
```

Step 4 — Inspect the Dataset

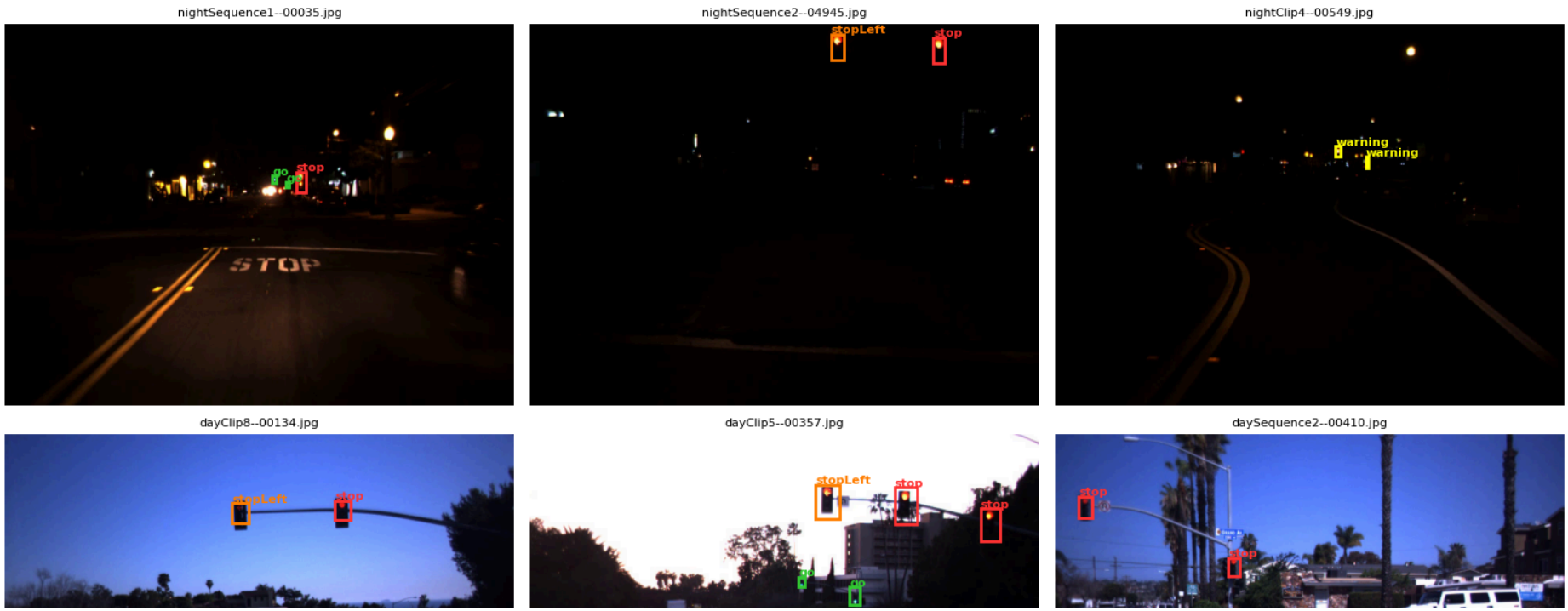
```
import sys
from pathlib import Path
from collections import Counter

# Ensure training modules are importable (Colab + local)
training_dir = REPO_ROOT / 'training'
if str(training_dir) not in sys.path:
    sys.path.insert(0, str(training_dir))

from lisa_classes import CLASS_NAMES

# ── File counts ──────────────────────────────────────────────────────
train_imgs = list((YOLO_DATA / 'images' / 'train').glob('*.jpg'))
val_imgs   = list((YOLO_DATA / 'images' / 'val').glob('*.jpg'))
print(f'Train images : {len(train_imgs)}')
print(f'Val   images : {len(val_imgs)}')

# ── Class distribution ───────────────────────────────────────────────
counter: Counter = Counter()
for split in ('train', 'val'):
    for lbl_file in (YOLO_DATA / 'labels' / split).glob('*.txt'):
        for line in lbl_file.read_text().splitlines():
            if line.strip():
                counter[int(line.split()[0])] += 1

print('\nAnnotation counts per class:')
for idx, name in enumerate(CLASS_NAMES):
    print(f'  {idx:2d}  {name:<20s} {counter.get(idx, 0):>6d}')
```

```
Train images : 30826
Val   images : 5439

Annotation counts per class:
   0  go                   46723
   1  goForward              205
   2  goLeft                2476
   3  stop                 44318
```

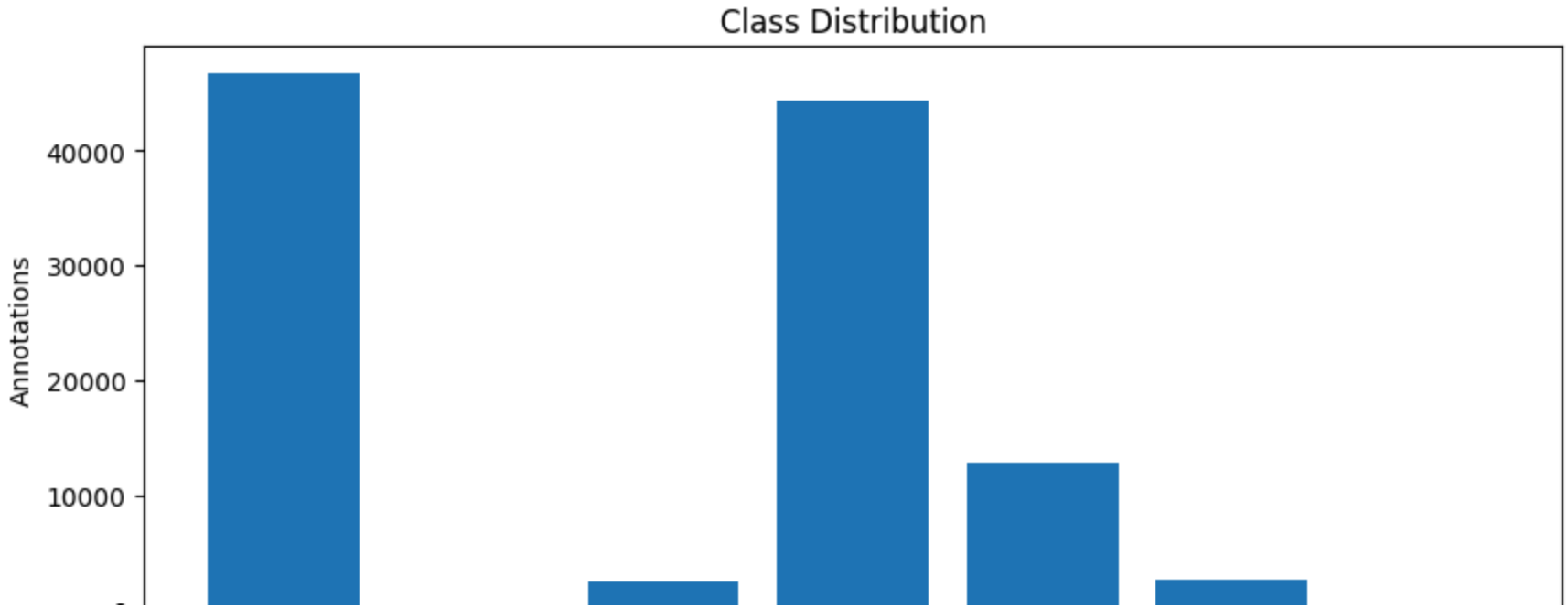
```
   4  stopLeft              12734
   5  warning                2669
   6  warningLeft             350
```

```
# ── Visualise a random training sample ──────────────────────────────────────
import random
import cv2
import matplotlib.pyplot as plt
import matplotlib.patches as patches

COLORS = [
    (0.2, 0.8, 0.2),  # go         — green
    (0.0, 0.6, 1.0),  # goForward  — blue
    (0.0, 1.0, 1.0),  # goLeft     — cyan
    (1.0, 0.2, 0.2),  # stop       — red
    (1.0, 0.5, 0.0),  # stopLeft   — orange
    (1.0, 1.0, 0.0),  # warning    — yellow
    (0.8, 0.0, 0.8),  # warningLeft— purple
]

random.seed(0)
sample_imgs = random.sample(train_imgs, min(6, len(train_imgs)))

fig, axes = plt.subplots(2, 3, figsize=(15, 8))
for ax, img_path in zip(axes.flat, sample_imgs):
    img = cv2.cvtColor(cv2.imread(str(img_path)), cv2.COLOR_BGR2RGB)
    h, w = img.shape[:2]
    ax.imshow(img)

    lbl_path = YOLO_DATA / "labels" / "train" / (img_path.stem + ".txt")
    if lbl_path.exists():
        for line in lbl_path.read_text().splitlines():
            parts = line.split()
            if len(parts) < 5:
                continue
            cls_idx = int(parts[0])
            cx, cy, bw, bh = map(float, parts[1:])
            x1 = (cx - bw / 2) * w
            y1 = (cy - bh / 2) * h
            color = COLORS[cls_idx % len(COLORS)]
```

```
            rect = patches.Rectangle(
                (x1, y1), bw * w, bh * h,
                linewidth=2, edgecolor=color, facecolor="none"
            )
            ax.add_patch(rect)
            ax.text(
                x1, y1 - 4, CLASS_NAMES[cls_idx],
                color=color, fontsize=8, fontweight="bold"
            )

    ax.set_title(img_path.name, fontsize=8)
    ax.axis("off")

plt.tight_layout()
plt.show()
```



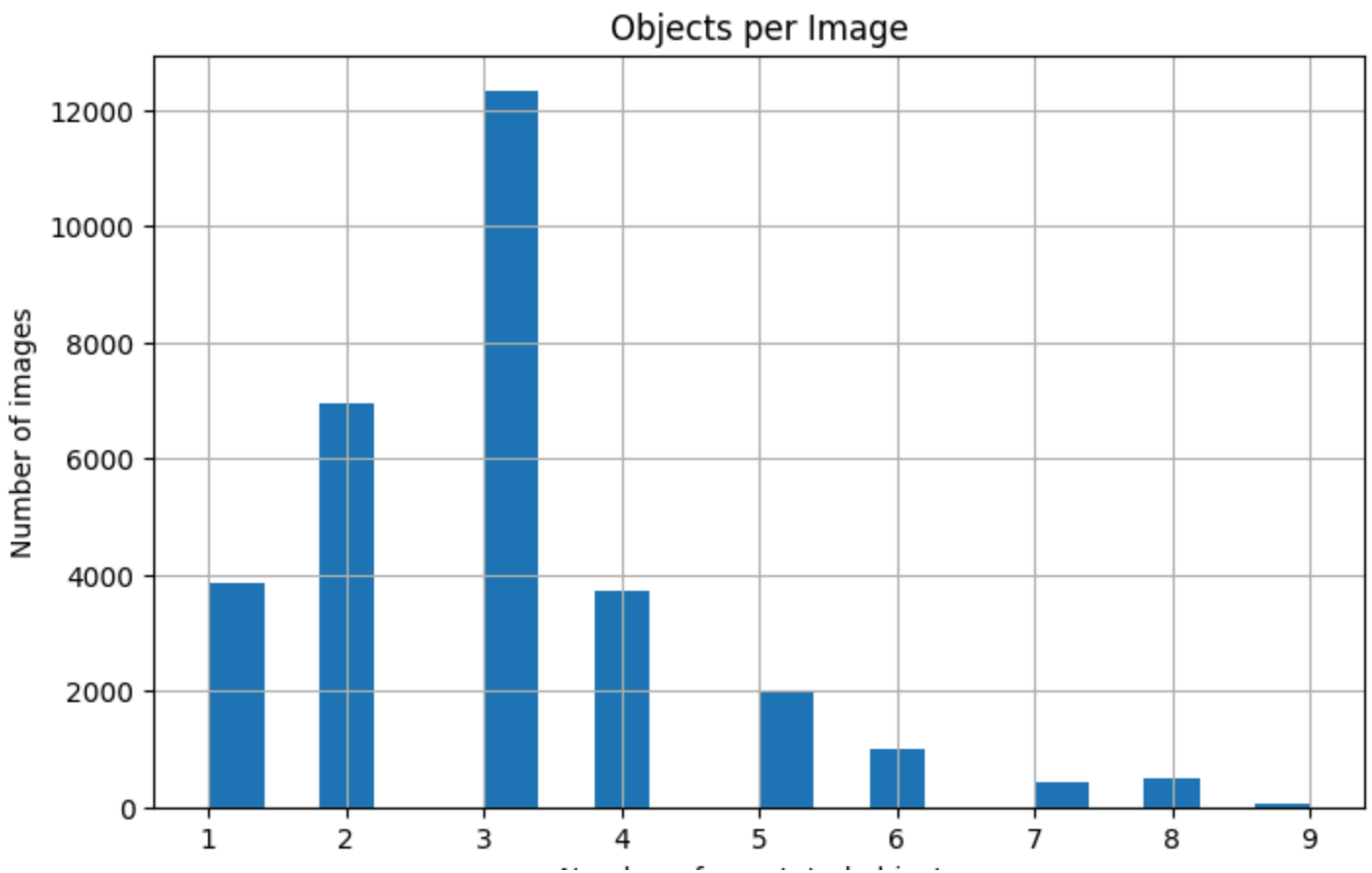
```
import matplotlib.pyplot as plt

classes = [
"go","goForward","goLeft",
"stop","stopLeft","warning","warningLeft"
]

counts = [46723,205,2476,44318,12734,2669,350]

plt.figure(figsize=(10,4))
plt.bar(classes, counts)
plt.xticks(rotation=45)
plt.title("Class Distribution")
plt.ylabel("Annotations")
plt.show()
```




```
# Count how many annotated objects appear in each training image
# and plot a histogram of objects per image

from pathlib import Path
import matplotlib.pyplot as plt

label_dir = YOLO_DATA / "labels" / "train"

objects_per_image = []

for label_file in label_dir.glob("*.txt"):
    with open(label_file, "r", encoding="utf-8") as f:
        lines = [line.strip() for line in f if line.strip()]
        objects_per_image.append(len(lines))

print("Total images checked:", len(objects_per_image))
```

```
print("Min objects in an image:", min(objects_per_image))
print("Max objects in an image:", max(objects_per_image))

plt.figure(figsize=(8, 5))
plt.hist(objects_per_image, bins=20)
plt.title("Objects per Image")
plt.xlabel("Number of annotated objects")
plt.ylabel("Number of images")
plt.grid(True)
plt.show()
```

```
Total images checked: 30826
Min objects in an image: 1
Max objects in an image: 9
```



```
# Determine which classes are rare based on annotation counts
# Classes below a chosen threshold will be treated as rare

class_counts = {
    "go": 46723,
    "goForward": 205,
    "goLeft": 2476,
    "stop": 44318,
    "stopLeft": 12734,
    "warning": 2669,
    "warningLeft": 350
}

threshold = 3000

rare_classes = [cls for cls, count in class_counts.items() if count < threshold]

print("Rare classes:", rare_classes)
```

```
Rare classes: ['goForward', 'goLeft', 'warning', 'warningLeft']
```

```
# Count train images and labels in the prepared dataset
train_images = list((YOLO_DATA / "images" / "train").glob("*"))
train_labels = list((YOLO_DATA / "labels" / "train").glob("*.txt"))

print("Train images:", len(train_images))
print("Train labels:", len(train_labels))
```

```
Train images: 30826
Train labels: 30826
```

```
# Count how many images contain each class
# This is often more useful than raw annotation totals for imbalance checks

from collections import Counter

label_dir = YOLO_DATA / "labels" / "train"

images_per_class = Counter()
```

```python
annotations_per_class = Counter()

for label_file in label_dir.glob("*.txt"):
    classes_in_image = set()

    with open(label_file, "r", encoding="utf-8") as f:
        for line in f:
            parts = line.strip().split()
            if not parts:
                continue

            class_id = int(parts[0])
            annotations_per_class[class_id] += 1
            classes_in_image.add(class_id)

    for class_id in classes_in_image:
        images_per_class[class_id] += 1

print("Images containing each class:")
for class_id, count in sorted(images_per_class.items()):
    print(class_id, count)

print("\nAnnotation counts per class:")
for class_id, count in sorted(annotations_per_class.items()):
    print(class_id, count)
```

```
Images containing each class:
0 15833
1 131
2 1590
3 15821
4 8818
5 933
6 198

Annotation counts per class:
0 39650
1 166
2 2115
3 37749
4 10817
```

```
5 2269
6 306
```

```
# Oversample rare-class images safely

# Import required modules
import shutil
from pathlib import Path

# Point to the prepared YOLO dataset folders
label_dir = YOLO_DATA / "labels" / "train"
image_dir = YOLO_DATA / "images" / "train"

print("Label dir:", label_dir)
print("Image dir:", image_dir)

# Define rare class IDs based on your analysis
rare_classes = [1, 6]   # goForward, warningLeft

oversample_factor = 3   # safer than 4 for large datasets

for label_file in label_dir.glob("*.txt"):

    # Skip files that are already augmented
    if "_aug" in label_file.stem:
        continue

    # Read classes present in this image
    with open(label_file) as f:
        classes = [int(line.split()[0]) for line in f.readlines()]

    # If image contains rare classes
    if any(c in rare_classes for c in classes):

        # Find corresponding image file
        img_candidates = list(image_dir.glob(label_file.stem + ".*"))
        if not img_candidates:
            continue

        img_file = img_candidates[0]
```

```
        # Create duplicated copies
        for i in range(oversample_factor):

            new_img = image_dir / f"{label_file.stem}_aug{i}{img_file.suffix}"
            new_lbl = label_dir / f"{label_file.stem}_aug{i}.txt"

            if not new_img.exists():
                shutil.copy(img_file, new_img)
                shutil.copy(label_file, new_lbl)
```

```
Label dir: /content/lisa_yolo/labels/train
Image dir: /content/lisa_yolo/images/train
```

```
# Count how many images contain each class after oversampling

from collections import Counter

label_dir = YOLO_DATA / "labels" / "train"

images_per_class = Counter()
annotations_per_class = Counter()

for label_file in label_dir.glob("*.txt"):
    classes_in_image = set()

    with open(label_file, "r", encoding="utf-8") as f:
        for line in f:
            parts = line.strip().split()
            if not parts:
                continue

            class_id = int(parts[0])
            annotations_per_class[class_id] += 1
            classes_in_image.add(class_id)

    for class_id in classes_in_image:
        images_per_class[class_id] += 1

print("Images containing each class:")
```

```
for class_id, count in sorted(images_per_class.items()):
    print(class_id, count)

print("\nAnnotation counts per class:")
for class_id, count in sorted(annotations_per_class.items()):
    print(class_id, count)
```

```
Images containing each class:
0 16262
1 524
2 1758
3 15821
4 8866
5 1215
6 792

Annotation counts per class:
0 40427
1 664
2 2283
3 37749
4 10865
5 3115
6 1224
```

Step 5 — Load the Base Model

```
from ultralytics import YOLO

model = YOLO(str(BASE_MODEL))
print("Model loaded:", BASE_MODEL.name)
print("Parameters  :", sum(p.numel() for p in model.model.parameters()), "total")
```

```
Model loaded: yolov8n_lisa_best.pt
Parameters  : 3012213 total
```

Step 6 — Train

Progress bars and per-epoch metrics will stream directly into this cell's output. Training writes checkpoints to `runs/train/lisa_yolov8n/weights/`.

```
from ultralytics import YOLO

# yaml_path may not be set if Step 3 was skipped — resolve from config
yaml_path = YOLO_DATA / 'dataset.yaml'
if not yaml_path.exists():
    raise FileNotFoundError(
        f'Dataset YAML not found: {yaml_path}\nRun Step 3 first.'
    )

model = YOLO(str(BASE_MODEL))

train_kwargs = dict(
    data       = str(yaml_path.resolve()),
    epochs     = EPOCHS,
    imgsz      = IMG_SIZE,
    batch      = BATCH,
    workers    = WORKERS,
    device     = DEVICE,
    lr0        = LR0,
    patience   = PATIENCE,
    project    = RUN_PROJECT,
    name       = RUN_NAME,
    exist_ok   = True,
    # Augmentation parameters
    mosaic     = 1.0 if AUGMENT else 0.0,
    mixup      = 0.1 if AUGMENT else 0.0,
    hsv_h      = 0.015 if AUGMENT else 0.0,
    hsv_s      = 0.7 if AUGMENT else 0.0,
    hsv_v      = 0.4 if AUGMENT else 0.0,
    degrees    = 10 if AUGMENT else 0.0,
    translate  = 0.1 if AUGMENT else 0.0,
    scale      = 0.5 if AUGMENT else 0.0,
    fliplr     = 0.5 if AUGMENT else 0.0,

    copy_paste = 0.0,   # disabled — causes shape mismatch on LISA
    freeze     = FREEZE if FREEZE > 0 else None,
```

```
    verbose    = True,
    save       = True,
    save_period= -1,
    plots      = False,
)

print('Training config:')
for k, v in train_kwargs.items():
    print(f'  {k:<12}: {v}')

results = model.train(**train_kwargs)
print('\nTraining complete.')
```

```
Training config:
  data        : /content/lisa_yolo/dataset.yaml
  epochs      : 50
  imgsz       : 640
  batch       : 64
  workers     : 8
  device      : 0
  lr0         : 0.001
  patience    : 20
  project     : /content/runs/train
  name        : lisa_yolov8n_v1.1
  exist_ok    : True
  mosaic      : 1.0
  mixup       : 0.1
  hsv_h       : 0.015
  hsv_s       : 0.7
  hsv_v       : 0.4
  degrees     : 10
  translate   : 0.1
  scale       : 0.5
  fliplr      : 0.5
  copy_paste  : 0.0
  freeze      : None
  verbose     : True
  save        : True
  save_period : -1
  plots       : False
New https://pypi.org/project/ultralytics/8.4.22 available 😃 Update with 'pip install -U ultralytics'
Ultralytics 8.4.21 🚀 Python-3.12.12 torch-2.10.0+cu128 CUDA:0 (NVIDIA H100 80GB HBM3, 81079MiB)
```

```
engine/trainer: agnostic_nms=False, amp=True, angle=1.0, augment=False, auto_augment=randaugment, batch=64, b

                   from  n    params  module                                       arguments
  0                  -1  1       464  ultralytics.nn.modules.conv.Conv             [3, 16, 3, 2]
  1                  -1  1      4672  ultralytics.nn.modules.conv.Conv             [16, 32, 3, 2]
  2                  -1  1      7360  ultralytics.nn.modules.block.C2f             [32, 32, 1, True]
  3                  -1  1     18560  ultralytics.nn.modules.conv.Conv             [32, 64, 3, 2]
  4                  -1  2     49664  ultralytics.nn.modules.block.C2f             [64, 64, 2, True]
  5                  -1  1     73984  ultralytics.nn.modules.conv.Conv             [64, 128, 3, 2]
  6                  -1  2    197632  ultralytics.nn.modules.block.C2f             [128, 128, 2, True]
  7                  -1  1    295424  ultralytics.nn.modules.conv.Conv             [128, 256, 3, 2]
  8                  -1  1    460288  ultralytics.nn.modules.block.C2f             [256, 256, 1, True]
  9                  -1  1    164608  ultralytics.nn.modules.block.SPPF            [256, 256, 5]
 10                  -1  1         0  torch.nn.modules.upsampling.Upsample         [None, 2, 'nearest']
 11             [-1, 6]  1         0  ultralytics.nn.modules.conv.Concat           [1]
 12                  -1  1    148224  ultralytics.nn.modules.block.C2f             [384, 128, 1]
 13                  -1  1         0  torch.nn.modules.upsampling.Upsample         [None, 2, 'nearest']
 14             [-1, 4]  1         0  ultralytics.nn.modules.conv.Concat           [1]
 15                  -1  1     37248  ultralytics.nn.modules.block.C2f             [192, 64, 1]
 16                  -1  1     36992  ultralytics.nn.modules.conv.Conv             [64, 64, 3, 2]
 17            [-1, 12]  1         0  ultralytics.nn.modules.conv.Concat           [1]
 18                  -1  1    123648  ultralytics.nn.modules.block.C2f             [192, 128, 1]
 19                  -1  1    147712  ultralytics.nn.modules.conv.Conv             [128, 128, 3, 2]
 20             [-1, 9]  1         0  ultralytics.nn.modules.conv.Concat           [1]
 21                  -1  1    493056  ultralytics.nn.modules.block.C2f             [384, 256, 1]
 22        [15, 18, 21]  1    752677  ultralytics.nn.modules.head.Detect           [7, 16, None, [64, 128, 25
Model summary: 130 layers, 3,012,213 parameters, 3,012,197 gradients, 8.2 GFLOPs
```

Step 7 — Review Metrics

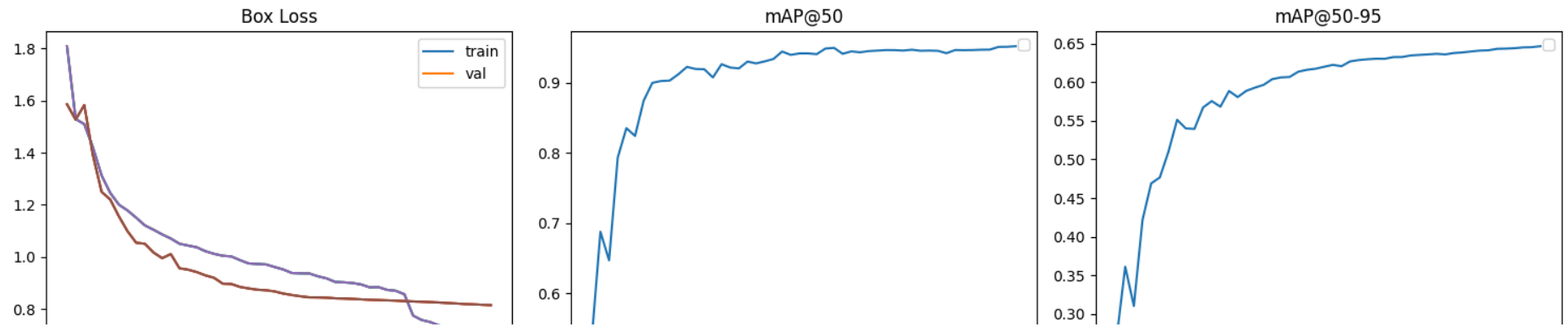
```
# ── Scalar summary ──────────────────────────────────────────────────────
metrics = results.results_dict
print("Final validation metrics:")
for key in (
    "metrics/mAP50(B)",
    "metrics/mAP50-95(B)",
    "metrics/precision(B)",
    "metrics/recall(B)",
):
    print(f"  {key:<30s}: {metrics.get(key, float('nan')):.4f}")
```

```
Final validation metrics:
  metrics/mAP50(B)              : 0.9519
  metrics/mAP50-95(B)           : 0.6469
  metrics/precision(B)          : 0.9283
  metrics/recall(B)             : 0.9221
```

```
# ── Training curves (loss + mAP) ────────────────────────────────────────
import pandas as pd
import matplotlib.pyplot as plt

results_csv = Path(RUN_PROJECT) / RUN_NAME / "results.csv"

if results_csv.exists():
    df = pd.read_csv(results_csv)
    df.columns = df.columns.str.strip()

    fig, axes = plt.subplots(1, 3, figsize=(15, 4))

    # Box loss
    for col, ax, title in [
        ("train/box_loss",        axes[0], "Box Loss"),
        ("metrics/mAP50(B)",      axes[1], "mAP@50"),
        ("metrics/mAP50-95(B)",   axes[2], "mAP@50-95"),
    ]:
        if col in df.columns:
            axes[0].plot(df["epoch"], df["train/box_loss"], label="train")
            if "val/box_loss" in df.columns:
                axes[0].plot(df["epoch"], df["val/box_loss"], label="val")
        ax.set_title(title)
```

```
        ax.set_xlabel("Epoch")
        ax.legend()

    if "metrics/mAP50(B)" in df.columns:
        axes[1].plot(df["epoch"], df["metrics/mAP50(B)"])
    if "metrics/mAP50-95(B)" in df.columns:
        axes[2].plot(df["epoch"], df["metrics/mAP50-95(B)"])

    plt.tight_layout()
    plt.show()
else:
    print("results.csv not found — check", RUN_PROJECT)
```

```
/tmp/ipykernel_860/150780517.py:25: UserWarning: No artists with labels found to put in legend.  Note that art
  ax.legend()
```



Step 8 — Save Best Model to `trained_models/`

```
import shutil

run_dir = Path(RUN_PROJECT) / RUN_NAME
best_pt = run_dir / "weights" / "best.pt"
last_pt = run_dir / "weights" / "last.pt"

OUT_DIR.mkdir(parents=True, exist_ok=True)

dest_best = OUT_DIR / "yolov8n_omniview_v1.1.pt"
shutil.copy2(best_pt, dest_best)
print("Best model  →", dest_best)

if last_pt.exists():
    dest_last = OUT_DIR / "yolov8n_lisa_last.pt"
    shutil.copy2(last_pt, dest_last)
    print("Last checkpoint →", dest_last)
```

```
Best model  → /content/drive/MyDrive/OmniAI/training_models/yolov8n_omniview_v1.1.pt
Last checkpoint → /content/drive/MyDrive/OmniAI/training_models/yolov8n_lisa_last.pt
```

Start coding or generate with AI.

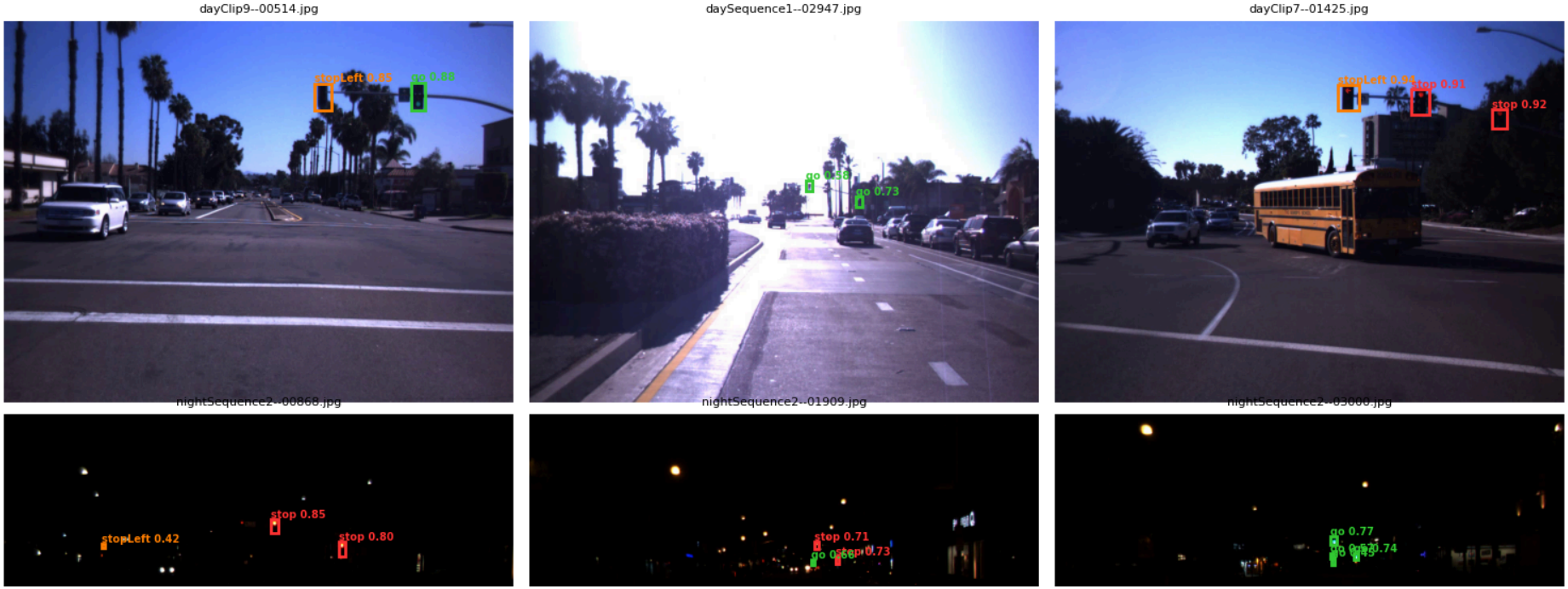
Step 9 — Quick Validation Inference

Runs the saved model on a few val images so you can visually confirm it is detecting signs.

```
import random
import cv2
import matplotlib.pyplot as plt
import matplotlib.patches as patches
from ultralytics import YOLO

infer_model = YOLO(str(dest_best))

val_imgs_list = list((YOLO_DATA / "images" / "val").glob("*.jpg"))
random.seed(1)
sample = random.sample(val_imgs_list, min(6, len(val_imgs_list)))
```

```
fig, axes = plt.subplots(2, 3, figsize=(15, 8))
for ax, img_path in zip(axes.flat, sample):
    img_bgr = cv2.imread(str(img_path))
    img_rgb = cv2.cvtColor(img_bgr, cv2.COLOR_BGR2RGB)
    h, w = img_rgb.shape[:2]

    preds = infer_model.predict(img_bgr, conf=0.25, verbose=False)[0]

    ax.imshow(img_rgb)
    for box in preds.boxes:
        x1, y1, x2, y2 = box.xyxy[0].tolist()
        cls_idx = int(box.cls[0])
        conf    = float(box.conf[0])
        color   = COLORS[cls_idx % len(COLORS)]
        rect = patches.Rectangle(
            (x1, y1), x2 - x1, y2 - y1,
            linewidth=2, edgecolor=color, facecolor="none"
        )
        ax.add_patch(rect)
        label = f"{CLASS_NAMES[cls_idx]} {conf:.2f}"
        ax.text(x1, y1 - 4, label, color=color, fontsize=7, fontweight="bold")

    ax.set_title(img_path.name, fontsize=8)
    ax.axis("off")

plt.suptitle("Fine-tuned YOLOv8n — Val Set Predictions", fontsize=12)
plt.tight_layout()
plt.show()
```

Fine-tuned YOLOv8n — Val Set Predictions



```
#  Blur license plates in test videos
# Add this after the quick validation inference cell.
# This assumes your model has a class for license plates.

import cv2
from ultralytics import YOLO
from pathlib import Path

video_model = YOLO(str(dest_best))

input_video = "/content/test_video.mp4"
output_video = "/content/test_video_blurred.mp4"

cap = cv2.VideoCapture(input_video)

fps = cap.get(cv2.CAP_PROP_FPS)
width = int(cap.get(cv2.CAP_PROP_FRAME_WIDTH))
height = int(cap.get(cv2.CAP_PROP_FRAME_HEIGHT))

writer = cv2.VideoWriter(
    output_video,
    cv2.VideoWriter_fourcc(*"mp4v"),
    fps,
    (width, height)
)

# Change this to the actual class name or class index used in your model
LICENSE_PLATE_CLASS_NAME = "license_plate"

while True:
    ret, frame = cap.read()
    if not ret:
        break

    result = video_model.predict(frame, conf=0.25, verbose=False)[0]

    for box in result.boxes:
        cls_idx = int(box.cls[0])
```

```
        cls_name = result.names[cls_idx]

        if cls_name == LICENSE_PLATE_CLASS_NAME:
            x1, y1, x2, y2 = map(int, box.xyxy[0].tolist())

            # Keep coordinates inside the frame
            x1 = max(0, x1)
            y1 = max(0, y1)
            x2 = min(frame.shape[1], x2)
            y2 = min(frame.shape[0], y2)

            roi = frame[y1:y2, x1:x2]
            if roi.size > 0:
                blurred_roi = cv2.GaussianBlur(roi, (51, 51), 0)
                frame[y1:y2, x1:x2] = blurred_roi

    writer.write(frame)

cap.release()
```